AI ASSISTANT CODING

ASSIGNMENT-16

Name: Kommu Madhupriya

Hallticket:2303A51583

Batch:22

Task Descrption-1: (Design Database Schema for Student Management System)

Instructions:

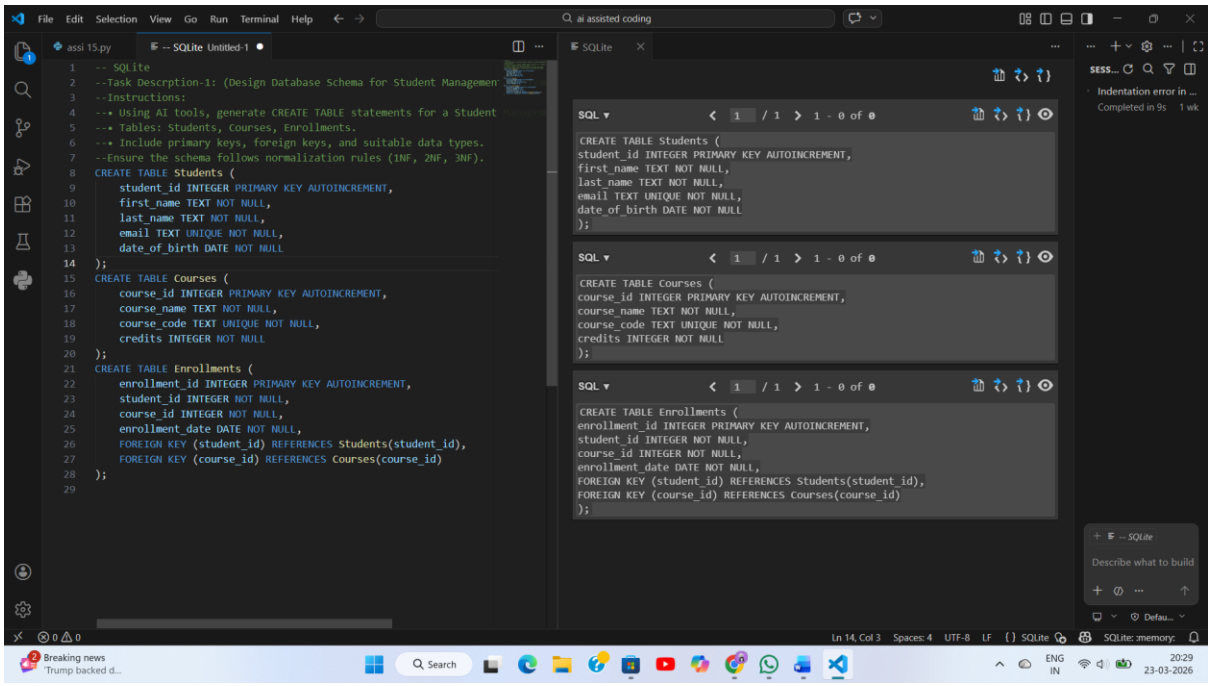
• Using AI tools, generate CREATE TABLE statements for a Student Management Database.

• Tables: Students, Courses, Enrollments.

• Include primary keys, foreign keys, and suitable data types.

Ensure the schema follows normalization rules (1NF, 2NF, 3NF).

Code:



Task Description- 2: (Insert Sample Records)

Instructions:

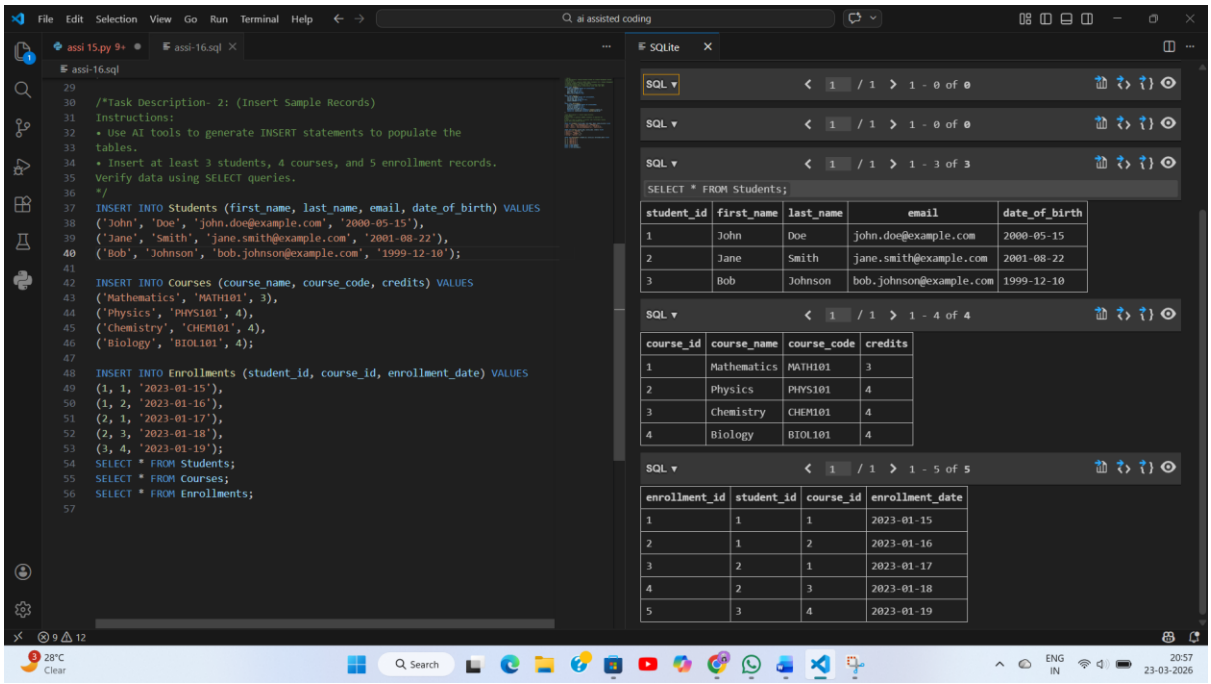
• Use AI tools to generate INSERT statements to populate the

tables.

• Insert at least 3 students, 4 courses, and 5 enrollment records.

Verify data using SELECT queries.

Code:



Task Description- 3: (Perform CRUD Operations)

Instructions:

Use AI to generate SQL queries for basic CRUD operations:

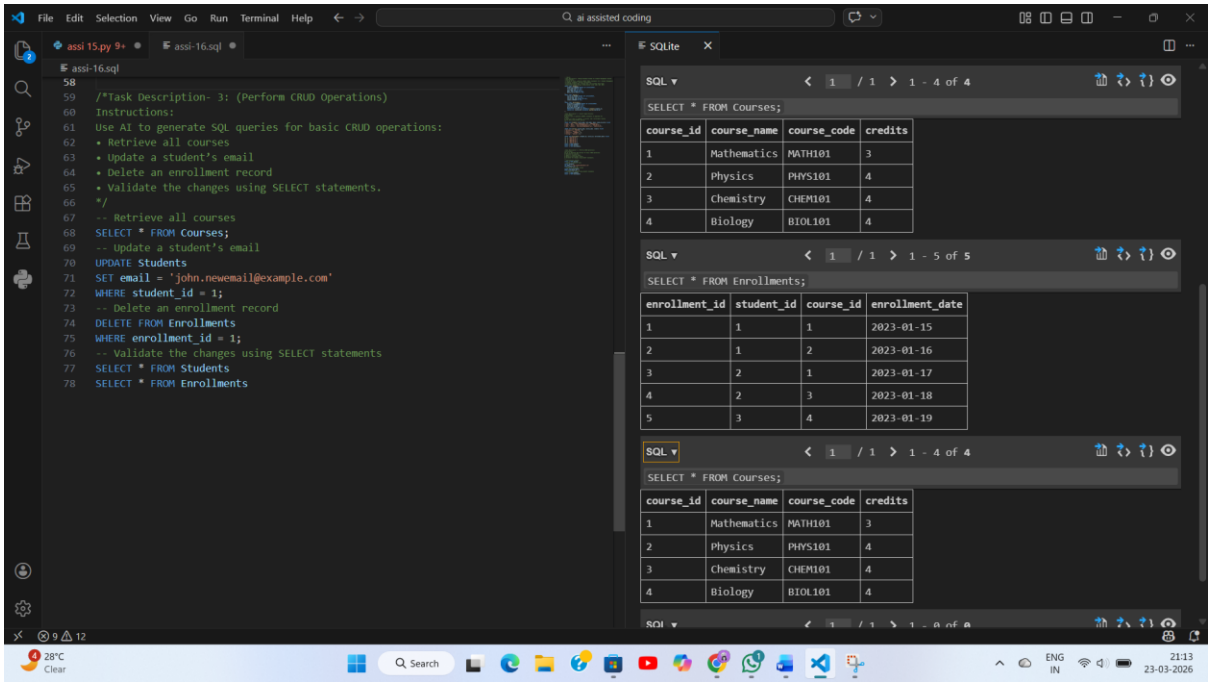
• Retrieve all courses

• Update a student's email

• Delete an enrollment record

• Validate the changes using SELECT statements.

Code:



Task Description- 4: (Execute Aggregate Queries)

Instructions:

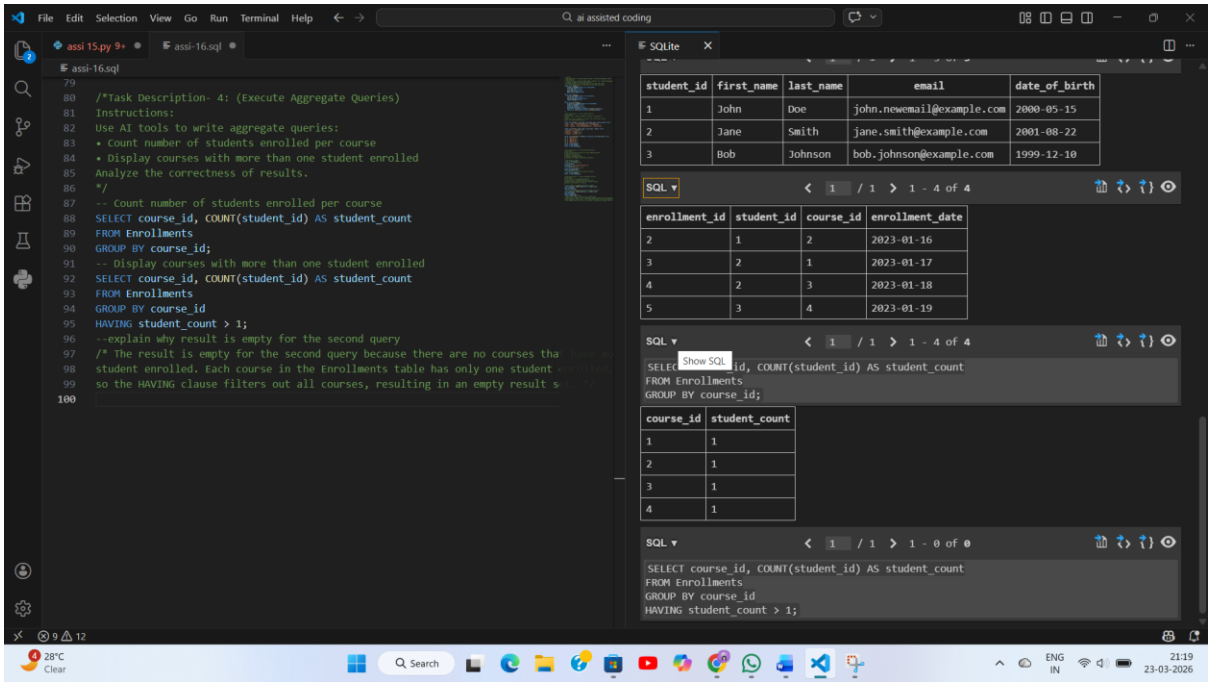
Use AI tools to write aggregate queries:

• Count number of students enrolled per course

• Display courses with more than one student enrolled

Analyze the correctness of results.

Code:



Task Description- 5: (Query Optimization)

Instructions:

• Use AI tools to analyze query performance.

• Rewrite inefficient sub queries using JOINs and test both queries

for correctness.

Code:

assi 15.py 9+ | assi-16.sql

assi-16.sql

```sql
/*Task Description- 5: (Query Optimization)
Instructions:
• Use AI tools to analyze query performance.
• Rewrite inefficient sub queries using JOINs and test both queries
for correctness.*/
-- Inefficient subquery using IN
SELECT course_name
FROM Courses
WHERE course_id IN (SELECT course_id FROM Enrollments WHERE student_id = 2);
-- Optimized query using JOIN
SELECT c.course_name
FROM Courses c
JOIN Enrollments e ON c.course_id = e.course_id
WHERE e.student_id = 1;
```

SQLite

3	2	1	2023-01-17
4	2	3	2023-01-18
5	3	4	2023-01-19

SQL ▾ 1 / 1 1 - 4 of 4

course_id	student_count
1	1
2	1
3	1
4	1

SQL ▾ 1 / 1 1 - 0 of 0

SQL ▾ 1 / 1 1 - 2 of 2

```sql
SELECT course_name
FROM Courses
WHERE course_id IN (SELECT course_id FROM Enrollments WHERE student_id = 2);
```

course_name
Mathematics
Chemistry

SQL ▾ 1 / 1 1 - 1 of 1

```sql
SELECT c.course_name
FROM Courses c
JOIN Enrollments e ON c.course_id = e.course_id
WHERE e.student_id = 1;
```

course_name
Physics